Case Study Report on

WELCOME APP

project submitted in fulfilment of the requirements for the

award of the degree of

BACHELOR OF TECHNOLOGY IN CSE (AI & ML)

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ANIL NEERUKONDA INSTITUTE OF TECHNOLOGY AND SCIENCES

(UGC AUTONOMOUS)

(Permanently Affiliated to AU, Approved by AICTE and Accredited by NBA

& NAAC with 'A+' Grade)

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BONAFIDE CERTIFICATE

This is to certify that the case study entitled "**WELCOME APP'** submitted by **A. Yaswanth Kumar, A. Reshma, B. Raghu Sasidhar, B. Sateesh Kumar , B. Jagadeesh, B. Jayanth , B.Pranathi Hiranmayi, B. Koushil** in fulfillment of the requirements for the award of Bachelor of Technology in CSE (AI & ML) in the department of Computer Science and Engineering (AI & ML) of Anil Neerukonda Institute of technology and sciences (A+), Visakhapatnam is a record of Bonafide work carried out under my guidance and supervision.

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ABSTRACT

This project presents an Android application designed to provide an engaging and interactive welcome experience for students entering the field of Artificial Intelligence and Machine Learning. The application focuses on delivering a visually appealing onboarding interface using modern UI design principles and smooth animations. Developed using Kotlin and XML in Android Studio, the app demonstrates the integration of dynamic content, user interaction, and aesthetic design elements.

The application opens with an animated welcome message, followed by a series of motivational AI-related messages that rotate automatically at regular intervals. It also includes an interactive button that allows users to manually explore content, enhancing user engagement. An expandable card provides detailed information about AI and ML, while a statistics section highlights key aspects in a visually attractive format. The design incorporates advanced concepts such as glassmorphism, neumorphism, and motion-based animations to create a futuristic user experience.

This project highlights the importance of intuitive design and interactive elements in mobile applications. It serves as a practical example of implementing Android development concepts such as layouts, event handling, animations, and responsive UI. Overall, the application demonstrates how technology can be used to create meaningful and engaging digital experiences for learners.

INTRODUCTION

Mobile applications have become an essential part of modern technology, providing interactive and user-friendly platforms for communication, learning, and innovation. In recent years, the field of Artificial Intelligence (AI) and Machine Learning (ML) has gained significant importance, influencing various industries such as healthcare, education, transportation, and finance. Introducing users to these advanced technologies in an engaging and understandable way is crucial, especially for students who are beginning their journey in AI and ML.

This project focuses on the development of an Android application that serves as an interactive welcome screen for AI & ML Engineering. The primary goal of the application is to create a visually appealing and immersive onboarding experience that captures the user's attention and encourages exploration. Unlike traditional static interfaces, this application integrates dynamic content, smooth animations, and modern design principles to enhance user interaction and engagement.

The application is developed using Kotlin as the programming language and XML for designing the user interface within Android Studio. It utilizes various UI components such as TextViews, Buttons, and Cards, combined with animation techniques like fade-in, slide transitions, and bounce effects. The app displays a welcome message followed by rotating motivational messages related to AI and ML concepts. Additionally, it includes an interactive button for manual navigation and an expandable card that provides detailed information about the field.

AIM

To design and develop an Android application that provides an interactive and visually engaging welcome interface for Artificial Intelligence and Machine Learning, using animations, dynamic content, and modern UI design principles to enhance user experience.

OBJECTIVE

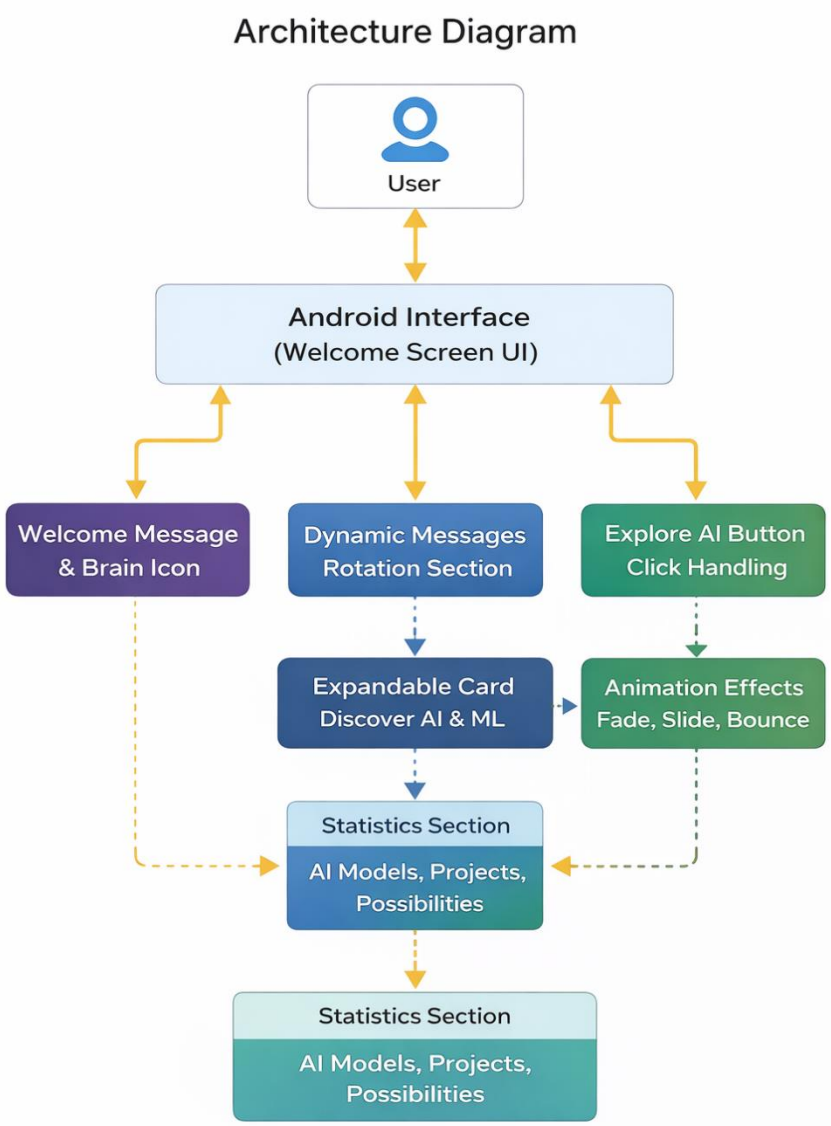
The primary objective of this case study is:

1. To design an interactive and visually appealing welcome screen.
2. To display dynamic AI & ML motivational messages.
3. To implement smooth animations for better user experience.
4. To handle user interactions using buttons and click events.
5. To create an expandable card for displaying detailed information.
6. To apply modern UI design techniques like glassmorphism and gradients.
7. To develop the application using Kotlin and XML in Android Studio.

TOOLS USED

Tool / Technology	Description
Android Studio	Integrated Development Environment used to build and run the Android application
Kotlin	Used for writing the application logic and handling interactions
XML	Used to design layouts, UI components, and resources
Material Design	Provides UI components and styling guidelines for modern app design
Android SDK 34	Platform used to develop and test the application on Android devices

ARCHITECTURE DIAGRAM



ARCHITECTURE EXPLANATION

The application follows a simple Android app architecture. The user interacts with the main welcome screen. The interface contains the welcome title, rotating motivational messages, an interactive button, an expandable information card, and a statistics section. The UI is designed using XML, while the app logic and event handling are implemented using Kotlin. Animations such as fade-in, slide, bounce, and rotation are applied to improve the visual experience and make the application more interactive.

PROCEDURE

- Install and open **Android Studio** on your system.
- Create a new Android project using **Kotlin** as the programming language.
- Select an Empty Activity and configure project details (name, package, SDK).
- Design the main UI layout using **XML** in the activity_main.xml file.
- Add UI components such as TextView, Button, ImageView, and CardView.
- Set the welcome message and AI brain icon in the layout.
- Implement rotating motivational messages using a timer or handler.
- Add animations like fade-in, slide-up, and rotation effects.
- Create the "Explore AI" button and implement click listener functionality.
- Add an expandable card to display AI & ML information.
- Design the statistics section with styled cards and values.
- Apply modern UI styles like gradients, glassmorphism, and color themes.
- Run the application on an emulator or Android device and verify the output.

IMPLEMENTATION

The Welcome Application was implemented in Android Studio using **Kotlin** for logic and **XML** for designing the user interface. The main screen contains a welcome title, rotating AI messages, an interactive button, an expandable information card, and a statistics section.

1. Main Activity Logic (Kotlin)

This file controls the rotating messages, button clicks, and expandable card behavior.

```
package com.example.welcomeapplication

import android.os.Bundle
import android.os.Handler
import android.os.Looper
import android.view.View
import android.widget.Button
import android.widget.ImageView
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity
import androidx.cardview.widget.CardView

class MainActivity : AppCompatActivity() {

    private lateinit var messageText: TextView
    private lateinit var counterText: TextView
```

```
private lateinit var exploreButton: Button
private lateinit var detailsCard: CardView
private lateinit var detailsText: TextView
private lateinit var brainIcon: ImageView

private val messages = listOf(
    "Build Intelligent Systems",
    "Explore Neural Networks",
    "Shape the Future with AI",
    "Master Deep Learning",
    "Innovate with Data Science",
    "Transform Ideas into Models"
)

private val buttonLabels = listOf(
    "Explore AI",
    "Next Insight",
    "Discover More",
    "Inspire Me"
)

private var currentMessageIndex = 0
private var currentButtonIndex = 0
private val handler = Handler(Looper.getMainLooper())

private val messageRunnable = object : Runnable {
    override fun run() {
        currentMessageIndex = (currentMessageIndex + 1) % messages.size
        updateMessage()
        handler.postDelayed(this, 3000)
    }
}

override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    setContentView(R.layout.activity_main)

    messageText = findViewById(R.id.messageText)
    counterText = findViewById(R.id.counterText)
    exploreButton = findViewById(R.id.exploreButton)
    detailsCard = findViewById(R.id.detailsCard)
    detailsText = findViewById(R.id.detailsText)
    brainIcon = findViewById(R.id.brainIcon)

    updateMessage()
```

```
        handler.postDelayed(messageRunnable, 3000)

        exploreButton.setOnClickListener {
            currentMessageIndex = (currentMessageIndex + 1) % messages.size
            currentButtonIndex = (currentButtonIndex + 1) % buttonLabels.size

            updateMessage()
            exploreButton.text = buttonLabels[currentButtonIndex]
        }

        detailsCard.setOnClickListener {
            if (detailsText.visibility == View.GONE) {
                detailsText.visibility = View.VISIBLE
            } else {
                detailsText.visibility = View.GONE
            }
        }
    }

    private fun updateMessage() {
        messageText.text = messages[currentMessageIndex]
        counterText.text = String.format("%02d / %02d", currentMessageIndex + 1,
messages.size)
    }

    override fun onDestroy() {
        super.onDestroy()
        handler.removeCallbacks(messageRunnable)
    }
}
```

2. XML Layout Design

This file defines the structure of the welcome screen.

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#0A0E27">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="24dp"
        android:gravity="center_horizontal">
```

```
<ImageView
    android:id="@+id/brainIcon"
    android:layout_width="100dp"
    android:layout_height="100dp"
    android:src="@drawable/ic_brain"
    android:contentDescription="AI Brain Icon"
    android:layout_marginTop="20dp" />

<TextView
    android:id="@+id/titleText"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Welcome to AI &amp; ML Engineering"
    android:textColor="#FFFFFF"
    android:textSize="26sp"
    android:textStyle="bold"
    android:gravity="center"
    android:layout_marginTop="20dp" />

<TextView
    android:id="@+id/messageText"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Build Intelligent Systems"
    android:textColor="#00F5D4"
    android:textSize="20sp"
    android:layout_marginTop="24dp"
    android:gravity="center" />

<TextView
    android:id="@+id/counterText"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="01 / 06"
    android:textColor="#BBBBBB"
    android:textSize="14sp"
    android:layout_marginTop="8dp" />

<Button
    android:id="@+id/exploreButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Explore AI"
    android:layout_marginTop="24dp"
    android:backgroundTint="#00BFFF"
```

```
        android:textColor="#FFFFFF" />

    <androidx.cardview.widget.CardView
        android:id="@+id/detailsCard"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="24dp"
        app:cardCornerRadius="16dp"
        app:cardElevation="8dp"
        app:cardBackgroundColor="#33FFFFFF">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="vertical"
            android:padding="16dp">

            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Discover AI &amp; ML"
                android:textColor="#FFFFFF"
                android:textSize="18sp"
                android:textStyle="bold" />

            <TextView
                android:id="@+id/detailsText"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:text="Artificial Intelligence and Machine Learning are transforming
every industry — from self-driving cars to medical diagnosis, AI is reshaping how we live
and work."
                android:textColor="#DDDDDD"
                android:textSize="16sp"
                android:visibility="gone"
                android:layout_marginTop="12dp" />
        </LinearLayout>

    </androidx.cardview.widget.CardView>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:layout_marginTop="24dp"
        android:gravity="center">
```

```
        <TextView
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:text="AI Models\n50+"
            android:textColor="#00BFFF"
            android:textStyle="bold"
            android:gravity="center"
            android:textSize="18sp" />

        <TextView
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:text="Projects\n100+"
            android:textColor="#8B5CF6"
            android:textStyle="bold"
            android:gravity="center"
            android:textSize="18sp" />

        <TextView
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:text="Possibilities\n∞"
            android:textColor="#00F5D4"
            android:textStyle="bold"
            android:gravity="center"
            android:textSize="18sp" />
    </LinearLayout>

  </LinearLayout>
</ScrollView>
```

3. Strings Resource File

This file stores reusable text values.

```
<resources>
  <string name="app_name">Welcome Application</string>
  <string name="welcome_title">Welcome to AI &amp; ML Engineering</string>
  <string name="explore_ai">Explore AI</string>
  <string name="discover_ai_ml">Discover AI &amp; ML</string>
</resources>
```

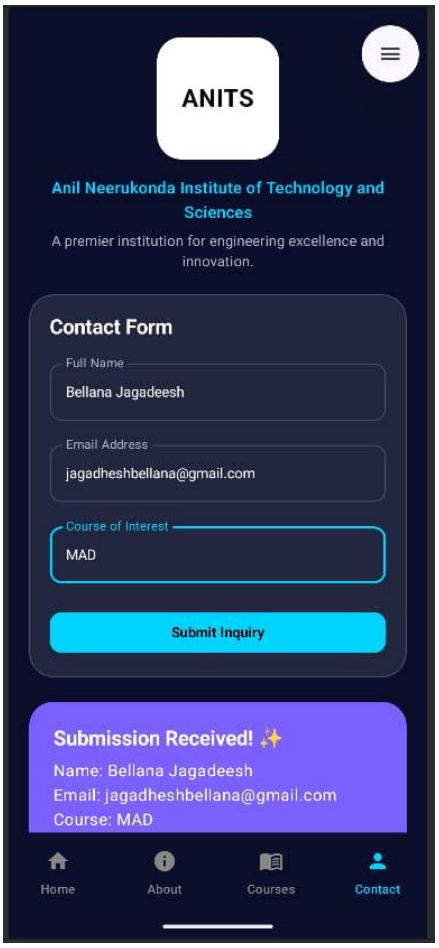
4. Explanation of Code

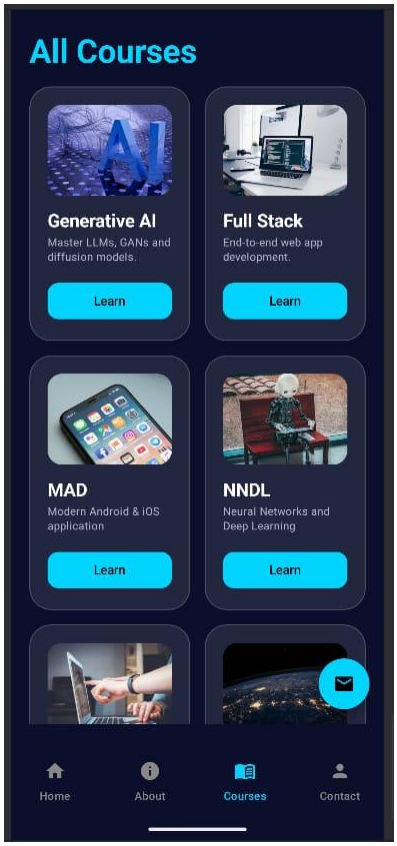
- **TextView** is used to display the welcome title, rotating messages, and counter.
- **Button** is used for manual message switching.
- **CardView** is used to create the expandable "Discover AI & ML" section.
- **Handler and Runnable** are used to automatically rotate messages every 3 seconds.
- **Click Listener** is used to respond when the user presses the button or card.
- **LinearLayout and ScrollView** are used to arrange all UI elements properly.

OUTPUT

After running the application:

- The welcome title is displayed.
- Messages rotate automatically.
- Clicking the button changes the message and button text.
- Clicking the card shows or hides detailed AI & ML information.
- Statistics are displayed at the bottom.

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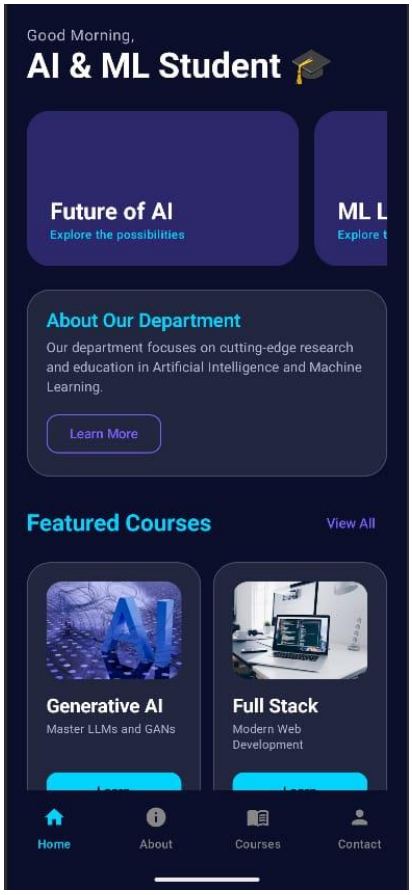
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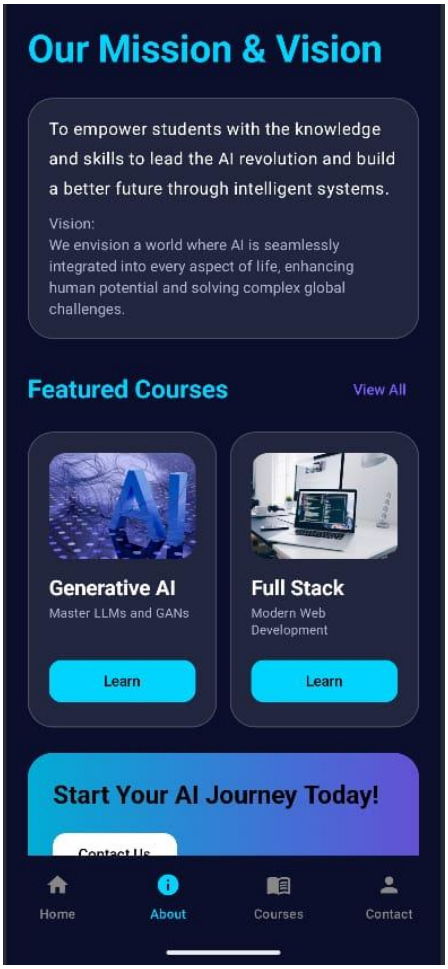

RESULTS

The Android application was successfully executed with an interactive welcome screen. Dynamic messages rotate automatically, and user interactions through the button and expandable card function correctly. All animations and UI components work smoothly. The app runs efficiently on devices, providing an engaging and visually appealing experience while meeting all project objectives.

CONCLUSION

This project successfully demonstrates the development of an interactive Android application that provides an engaging welcome experience for Artificial Intelligence and Machine Learning. By integrating dynamic content, smooth animations, and modern UI design, the application creates a visually appealing and user-friendly interface.

The use of Kotlin and XML helped in effectively implementing core Android concepts such as layout design, event handling, and animations. Features like rotating messages, interactive buttons, and expandable cards enhance user interaction and make the application more engaging.

The project highlights the importance of combining functionality with design to improve user experience. It also provides practical knowledge of Android development and showcases how applications can present complex topics like AI and ML in a simple and attractive manner.

ADVANTAGES

- Provides an engaging and interactive user experience
- Uses modern UI design techniques like animations and gradients
- Enhances understanding of Android development concepts
- Simple and easy-to-use interface
- Improves user engagement with dynamic content
- Lightweight application with smooth performance
- Can be extended for educational and real-world applications

LIMITATIONS

- Limited to basic functionality without advanced features
- No backend integration for data storage or processing
- Does not support user login or personalization
- Runs only on Android devices (no cross-platform support)
- Static content with no real-time data updates
- Not scalable for large or complex applications
- Limited error handling and validation
- Depends on device performance for animations
- No offline data saving or synchronization features

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